Francisco J. Campos-Arguedas

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Education

Ph.D. candidate in Horticulture **2026 (*expected*)**
University of Wisconsin–Madison
Dissertation: Effects of whole ecosystem warming on chill accumulation and cold hardiness of woody perennials.
Advisor: Dr. Al Kovaleski

M.S. in Plant Biology (Biologie végétale) **2021**
Université Laval (Québec, Canada)
Dissertation: Temperature effects on berry maturation and volatile compounds in *Vitis* sp. cv L'Acadie blanc during ripening.
Advisors: Dr. Martine Dorais and Dr. Karine Pedneault

B.S. in Agricultural Sciences **2017**
EARTH University (Limón, Costa Rica)
Dissertation: Development and production of an organomineral fertilizer in pellets and its field and laboratory evaluation.
Advisors: Dr. B.K. Singh and BSc. Ricardo Ulate Molina

Research and Professional Experience

Graduate Research Assistant **2021-Present**
Department of Plant and Agroecosystem Sciences, University of Wisconsin–Madison, USA

- Designed and conducted controlled-environment experiments to evaluate plant cold hardiness, chill accumulation, deacclimation, and budbreak.
- Performed growth chamber assays and managed experimental protocols.
- Prepared plant samples for Differential Thermal Analysis (DTA).
- Developed plant phenology models using R.
- Conducted comprehensive literature reviews and contributed to the preparation of scientific reports and manuscripts for publication.

Graduate Research Assistant **2019-2021**
Département de phytologie, Université Laval, Québec, Canada
Science department, Université Sainte-Anne, Nova Scotia, Canada

- Prepared samples and performed data analysis for plant and food-related research projects.
- Conducted GC–MS chromatogram integration and analysis.
- Collaborated in experimental winemaking processes and interdisciplinary research activities.
- Conducted literature reviews and contributed to scientific reports and manuscript preparation.

Civil Service Professional in Crop Sciences **2019**
National Institute of Innovation and Transfer in Agricultural Technology, Costa Rica.

- Worked as an Agronomic Engineer on the Soil Mapping Project.
- Performed soil physical analyses, including texture, bulk density, organic matter, hydraulic conductivity, moisture retention, pH, and acidity.
- Prepared soil samples for chemical analysis and laboratory processing.

Sustainability Program Coordinator **2018**
Buena Vista Lodge and Adventure Center, Costa Rica.

- Coordinated development and maintenance of vegetable gardens and biodigesters.
- Supervised organic and general waste collection programs.
- Supervised operations of the cosmetic laboratory.
- Managed and maintained compliance with the Certification for Sustainable Tourism (CST).

Experimental Organic Cocoa Plantation Supervisor **2017**
Integrated organic farm, EARTH University, Costa Rica
- Managed integrated production of an experimental organic cocoa plantation.
- Trained and supervised students in agricultural field practices.
- Collected and analyzed data on plant health, disease incidence, growth, and crop load.

Viticulture Lab Intern **2016**
Department of Viticulture and Oenology, Stellenbosch University, South Africa
- Conducted practical work in an experimental vineyard, including canopy management and pruning.
- Designed and implemented a plant phenology monitoring program.
- Performed HPLC analyses to quantify sugars and organic acids in grape berries.
- Participated in experimental wine production, from grape processing through fermentation.

Edible Mushroom Production Entrepreneurial Project **2012-2020**
- Developed and managed a family-owned oyster mushroom (*Pleurotus ostreatus*) production project.
- Oversaw the entire production process, from substrate preparation to harvesting and distribution.

Teaching & Mentorship Experience

Posse Program Mentor **2024-present**
The Posse Foundation
Division for Teaching and Learning, University of Wisconsin–Madison, USA
- Provide individual and group mentorship to Posse Scholars, supporting them through academic and personal challenges.
- Connect students with campus resources to foster academic success and leadership development.
- Collaborate with Posse staff, university liaisons, and students to promote a supportive and inclusive learning environment.

Principal Instructor **Fall 2024, Spring 2025, Fall 2025, Spring 2026**
CounPsy 115: Academic Excellence Seminar
Division for Teaching and Learning, University of Wisconsin–Madison.
- Designed and delivered a seminar supporting student persistence and academic skill development among Posse Scholars.
- Planned and facilitated weekly classes and active-learning activities.
- Managed course materials and communication through Canvas (learning management system).
- Developed group projects and individual assessments, including homework assignments.

Guest Lecture **Spring 2025**
Hort 330: Wines and Vines of the World
Department of Plant and Agroecosystem Sciences, University of Wisconsin–Madison
- Delivered a guest lecture on South African wine production, integrating viticultural, oenological, and regional perspectives.

Program Assistant **2023 & 2024**
Tropical Horticultural Systems - International Field Study in Costa Rica.
Department of Plant and Agroecosystem Sciences, University of Wisconsin–Madison
- Assisted with daily program coordination and field-based activities.
- Co-led evening reflection sessions incorporating active-learning and problem-based approaches.
- Facilitated communication among students, program leaders, and community partners.

Teaching Assistant **Spring 2023**
Hort 330: Wines and Vines of the World
Department of Plant and Agroecosystem Sciences, University of Wisconsin–Madison
- Prepared quizzes and assessments to evaluate student learning.
- Supported effective communication between students and the course instructor.
- Coordinated preparation and logistics for weekly wine tasting sessions.

Graduate Teaching Assistant **Fall 2019**
Biol 6013 Special topic
Science department, Université Sainte-Anne, Nova Scotia, Canada

- Trained and supervised students in laboratory techniques, material handling, and extractions.
- Planned laboratory activities and assigned tasks to support hands-on learning objectives.

Scientific dissemination

Publications:
**Denotes equal contribution*

- **Campos-Arguedas, F.***, Kirchhof, E.*, North, M.G., Londo, J.P., Bates, T., van Leeuwen, C., Destrac-Irvine, A., Bois, B., Kovaleski, A.P. (2026). Cold hardiness dynamics predict budbreak and associated low-temperature threats in grapevine. New Phytologist. https://doi.org/10.1111/nph.71074
- Kirchhof, E.*, **Campos-Arguedas, F.*,** Arias, N., Kovaleski, A.P. (2025). Thresholds for spring freeze: measuring risk to improve predictions in a warming world. New Phytologist. https://doi.org/10.1111/nph.70453
- Pedneault, K., Pico Carbajo, J., Nicolle, P., Pathak, M., Wilson, N., **Campos-Arguedas, F.**, Sarraihlé, G., Dorais, M., Brereton, N.J.B., Castellarin, S. and Pitre, F.E. (2023). Different patterns of temperature impact phenolic derivatives in ripening *Vitis* sp. berries. Acta Hortic. 1370, 61-66. doi: 10.17660/ActaHortic.2023.1370.9
- **Campos-Arguedas, F.**, Sarraihlé, G., Nicolle, P., Dorais, M. Brereton, NJB., Pitre, FE., Pedneault, K. 2022. Different Temperature and UV Patterns Modulate Berry Maturation and Volatile Compounds Accumulation in *Vitis* sp. Front. Plant Sci. 13:862259. doi: 10.3389/fpls.2022.862259

In preparation:

- **Campos-Arguedas, F.**, North, M.G., Kovaleski, A.P. Warming Reduces Plant Cold Hardiness but Cold Damage Risk Varies by Species and Season
- **Campos-Arguedas, F.**, North, M.G., Kovaleski, A.P. Exploring Dormancy: Responses to Whole Ecosystem Warming and Elevated CO_2 in a Boreal Peatland Forest
- **Campos-Arguedas, F.**, North, M.G., Kovaleski, A.P. Cold Hardiness Dynamics Predict Budbreak and Associated Low Temperature Threats in Boreal Ecosystems
- **Campos-Arguedas, F.**, North, M.G., Kovaleski, A.P. Supercool winter physiology: at what temperature do trees chill?

Presentations:

- **Campos-Arguedas, F.**, Kovaleski, A. 2026. Supercool winter physiology: at what temperature do trees chill? Presented at the 2026 UW–Madison Arboretum Research Symposium. Madison, WI. Feb 5, 2026.
- **Campos-Arguedas, F.**, Kovaleski, A. 2025. Ecosystem Warming May Increase or Decrease Probability of Low-Temperature Damage Depending on Tree Species. Presented at the 2025 SPRUCE All Hands On Meeting. Minneapolis, Minnesota. May 13-14, 2025.
- **Campos-Arguedas, F.**, North, M., Kovaleski, A. 2024. *Welcome to a Warmer Future*: Exploring Cold Hardiness and Dormancy Responses in Boreal Forest Species. Presented at 14th Annual Plant Sciences Symposium: "Growing into New Tech: Plant Sciences in Bloom". Madison, WI. Nov 15, 2024.
- **Campos-Arguedas, F.**, North, M., Kovaleski, A. 2023. Effects of Whole Ecosystem Warming on Cold Hardiness of Woody Perennials. Presented at the 2023 SPRUCE All Hands On Meeting. Minneapolis, Minnesota. May 2, 2023.
- **Campos-Arguedas, F.**, Sarraihlé, G., Dorais, M. Brereton, N., Pitre, F., Pedneault, K. 2021. Mild Temperature Increases During Early Growth Stages Impacts Benzene Derivatives Content in *Vitis* sp cv. L'Acadie Blanc Berries at Harvest. Presented at the 45th American Society for Enology and Viticulture-Eastern Section (ASEV-ES) Virtual Annual Meeting. Virtual event (via zoom): July 8, 2021.

Abstracts and posters:

- **Campos-Arguedas, F.**, North, M., Kovaleski, A. 2025. Species-Specific Responses to Ecosystem Warming May Lead to Greater or Lesser Vulnerability to Cold Damage American Geophysical Union. New Orleans, LA, USA. Dec 15-19, 2025.
- **Campos-Arguedas, F.**, Kirchhof, E., North, M., Kovaleski, A. 2025. Supercool winter physiology: at what temperature do trees chill? Arboretum Research Symposium poster session. Madison, Wisconsin, USA. Feb 6, 2025.

- **Campos-Arguedas, F.**, North, M., Kovaleski, A. 2024. Exploring Cold Hardiness and Dormancy: Boreal Forest Responses to Whole Ecosystem Warming. 14th Annual Plant Sciences Symposium poster session. UW–Madison. Madison, WI. Nov15, 2024.
- **Campos-Arguedas, F.**, North, M., Kovaleski, A. 2024. Exploring Cold Hardiness and Dormancy: Boreal Forest Responses to Whole Ecosystem Warming. 13th International Cold Hardiness Seminar (IPCHS) poster session. Clermont-Ferrand, France. Aug 26-30, 2024.
- **Campos-Arguedas, F.**, North, M., Kovaleski, A. 2024. Effects of Whole Ecosystem Warming on Cold Hardiness and Dormancy of Boreal Woody Perennial Plants. 12th annual Reid Bryson Scholarship poster session. Madison, WI, USA. Feb 26, 2024.
- **Campos-Arguedas, F.**, North, M., Kovaleski, A. 2022. Effects of Whole Ecosystem Warming on Cold Hardiness and Dormancy of Boreal Woody Perennial Plants. American Geophysical Union. Chicago, IL, USA. Dec 12-16, 2022.
- **Campos-Arguedas, F.**, Sarraihlé, G., Dorais, M. Brereton, N., Pitre, F., Pedneault, K. 2021. Temperature Spikes at Different Phenological Stages Impacts Biosynthesis of Volatile Compounds in *Vitis* sp. 'L'Acadie Blanc Berries Grown in Eastern Canada. 2021 Plant Biology Worldwide Summit. Virtual event: Jul 19-23, 2021.
- **Campos-Arguedas, F.**, Sarraihlé, G., Nicolle, P., Dorais, M., Brereton, N., Pitre, F., Pedneault, K. Impact de la hausse de température à différents moments pendant la saison de croissance sur le profil des composés volatiles glycosylés du cépage l'Acadie Blanc (*Vitis* sp.) *c*ultivé en Nouvelle-Écosse. Colloque des jeunes chercheuses et chercheurs, Université de Moncton. Virtual event: Mar 22-26, 2021.
- **Campos-Arguedas, F.**, Sarraihlé, G., Nicolle, P., Dorais, M., Brereton, N., Pitre, F., Pedneault, K. Impact de la hausse ponctuelle de température sur le profil des composés volatils glyscosylés du cépage L'Acadie Blanc (*Vitis* sp.) cultivé en Nouvelle-Écosse. Student scientific poster competition of the Center for Inter-institutional Research in Plant Sciences. Virtual event: Nov 25-26, 2020.
- **Campos-Arguedas, F.**, Sarraihlé, G., Nicolle, P., Dorais, M., Brereton, N., Pitre, F., Pedneault, K. Impact of mild temperature increases on the accumulation of bound volatile compounds in Vitis sp. L'acadie Blanc grown in eastern Canada. Plant Biology Worldwide Summit. DOI: 10.46678/PB.20.1052962. Virtual event: Jul 27-31, 2020.

Research Grants

- 2026-2028, National Institute of Food and Agriculture. Agriculture and Food Research Initiative Competitive Grants Program Foundational and Applied Science Program. *CranFROST: the Cranberry Freeze Risk and Outlook Support Tool.* PD: Al Kovaleski; Co-PD: Amaya Atucha, Mallika Nocco, Christopher Kucharik Graduate student: **Francisco Campos-Arguedas**. (*Pending $300,000*). Description: I contributed to the development and writing of the introduction, rationale and significance, methodology, and objectives sections of the proposal.
- 2026-2028, National Institute of Food and Agriculture. Agriculture and Food Research Initiative Competitive Grants Program Foundational and Applied Science Program. *Too cool for you? Measuring freeze risk threshold shifts with crop phenological progression.* PD: Al Kovaleski; Graduate student: **Francisco Campos-Arguedas**. (*Pending $299,984)*. Description: I contributed to the development and writing of the entire proposal.
- 2023-2025, University of Wisconsin-Madison Arboretum. *Supercool winter physiology: at what temperature do trees chill?* PI: Al Kovaleski; Graduate student: **Francisco Campos-Arguedas.** (*Funded $8000*). Description: I contributed to the development and writing of the entire proposal.

Leadership

Equipping Next Gen Ag and Conservation Professionals Mentorship Program 2024-2025

- Selected as a member and mentee in the Fall 2024 Next Generation Ag Leaders cohort.
- Participated in a leadership development program focused on agriculture, conservation, sustainability, and inclusion.
- Engaged with peer graduate students and community leaders to strengthen professional networks and contribute to equitable and sustainable agricultural systems.

Mentorship Opportunities in Science & Agriculture for Individuals of Color (MOSAIC) 2022-2025

- Served on the administrative board, supporting program planning and strategic direction.
- Planned and coordinated community-building events across the UW–Madison campus.

- Oversaw logistical planning, including scheduling, budgeting, and venue coordination.
- Contributed to grant writing efforts to support program sustainability and growth.

Plant Sciences Graduate Student Council (PSGSC) **2022**
- Served as Horticulture Representative for the 2022 term.
- Assisted in coordinating the 12th Plant Sciences Symposium, "*Cultivating Resilience: Learning from the Past to Plant for the Future*".
- Planned and organized monthly social and professional development activities.
- Acted as a point of contact between horticulture graduate students and the council.

EARTH Multicultural Fair **2017**
- Served as a member of the central organizing committee.
- Coordinated the events section, including cultural, musical, and entertainment programming.
- Managed artist outreach, contracting, and negotiations.
- Coordinated student schedules to ensure smooth execution of event activities.

Continuing Education and Professional Development

Spectral Ecology (SPEC) Summer School, Michigan State University **2026**
Training in ecological remote sensing (field and imaging spectroscopy, lidar, NEON data) and geospatial analysis; hybrid format (online and in-person)

International Professional Certificate: Profitable and Resilient Cacao, CATIE **Aug–Nov 2025**
Training in sustainable cacao production, agroforestry systems, and climate-resilient agricultural practices; online (120 hours)

Posse Foundation Local Mentor Training, UW–Madison Posse Team **Aug 2025**
Training in inclusive mentorship, student support, and leadership development; Madison, WI (20 hours)

Gender & Inclusion in Forest Landscape Restoration, CIFOR-ICRAF & Global Landscapes Forum **2025**
Training in gender equity, inclusion, and social dimensions of forest restoration; online (8 hours)

Research Mentor Training Program, Delta Program & UW–Madison Graduate School **Sep–Nov 2024**
Training in evidence-based mentoring practices, inclusive research environments, and student development; Madison, WI (15 hours)

Lecture Series for Young Scientists, 13th International Plant Cold Hardiness Seminar, INRAE **Aug 2024**
Advanced training in plant cold hardiness, stress physiology, and climate impacts on plant systems; Clermont-Ferrand, France (40 hours)

Posse Foundation Local Mentor Training, UW–Madison Posse Team **Aug 2024**
Training in mentorship, student engagement, and leadership in higher education; Madison, WI (20 hours)

Posse Foundation National Mentor Training, The Posse Foundation **Jun 2024**
National-level training in student mentorship, leadership development, and inclusive educational practices; New York, NY (28 hours)

Becoming a Resilient Scientist Workshop, UW–Madison Graduate School **May–Jun 2024**
Training in professional development, resilience, and well-being in academic research environments; Madison, WI (15 hours)

Family Farming in Latin America and the Caribbean, FAO **Apr 2018**
Training in food security, smallholder agriculture, and sustainable development in Latin America; online (80 hours)

Awards, Fellowships and Honors

CALS Graduate Mentoring Award **2026**
New Phytologist Foundation Poster Award - 13th International Cold Hardiness Seminar **2024**
Research Fellowship Award - University of Wisconsin–Madison Arboretum ($8000) **2024**
Bryson Scholarship - 12th annual Reid Bryson Scholarship poster session **2024**
Outstanding Student Presentation Award - American Geophysical Union **2022**
Forest Ecophysiology Poster Presentation Award - American Geophysical Union. **2022**
Student Research Grant Competition - University of Wisconsin–Madison **2022**
Aldo Leopold (Green Tree Garden Club) Scholarship - University of Wisconsin–Madison **2022**

Honor List (Awarded for academic excellence) - Université Laval **2021**
Best scientific poster - Center for Inter-institutional Research in Plant Sciences, Québec, Canada **2021**
Best presentation - Crops, Nutrition and Plant Health of the XV Graduation Seminar. EARTH University, Limón, Costa Rica **2017**
Distinguished student - EARTH University, Limón, Costa Rica **2015, 2016, 2017**
Academic honor - EARTH University, Limón, Costa Rica **2014, 2015, 2016**

Languages

- Spanish (native)
- English (advanced; TOEFL iBT)
- French (advanced)